# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **Triangles** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** In $\triangle ABC$ right-angled at $B$, $AB = 5$, $BC = 12$. If $BD \perp AC$, then $AD$ (in cm) is: 1

**(a)** $\frac{144}{13}$ **(b)** $\frac{5}{2}$ **(c)** $\frac{25}{13}$ **(d)** $\frac{60}{13}$

**2.** A square has diagonal $7\sqrt{2}$ cm. Its area (cm²) is: 1

**(a)** 98 **(b)** 196 **(c)** 49 **(d)** 14

**3.** In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 3 : 4$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 9: 1

**(a)** 40 **(b)** 16 **(c)** 58 **(d)** 49

**4.** In a right triangle with hypotenuse $c = 17$ and one leg $a = 8$, the area (sq. units) is: 1

**(a)** 120 **(b)** 60 **(c)** $\frac{255}{2}$ **(d)** 68

**5.** The altitude on the hypotenuse of the right triangle with legs 8 and 15 is of length (cm): 1

**(a)** $\frac{17}{2}$ **(b)** $\frac{120}{17}$ **(c)** 60 **(d)** $\frac{23}{2}$

**6.** If $\triangle ABC \sim \triangle PQR$, $ar(ABC) = 20$ and $ar(PQR) = 605$ and $AB = 4$, then $PQ =$ 1

**(a)** 24 **(b)** 4 **(c)** 22 **(d)** 242

**7.** In $\triangle ABC$ right-angled at $B$, $AB = 8$, $BC = 15$. If $BD \perp AC$, then $AD$ (in cm) is: 1

**(a)** $\frac{64}{17}$ **(b)** 4 **(c)** $\frac{120}{17}$ **(d)** $\frac{225}{17}$

**8.** A square has diagonal $9\sqrt{2}$ cm. Its area (cm²) is: 1

**(a)** 18 **(b)** 324 **(c)** 81 **(d)** 162

**9.** **Assertion (A):** A triangle with sides 5 cm, 6 cm and 7 cm is right-angled. 1
**Reason (R):** If in a triangle the square of one side equals the sum of the squares of the other two sides, the angle opposite the first side is a right angle.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** **Assertion (A):** The triangles with sides 3, 4, 5 and 6, 8, 10 are similar. 1
**Reason (R):** The hypotenuse is the longest side of a right-angled triangle.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** The areas of two similar triangles are $32$ cm² and $162$ cm². If a side of the first is 4 cm, find the corresponding side of the second. **2**

**12.** Find the altitude on the hypotenuse of the right triangle with legs 5 cm and 12 cm. **2**

**13.** In $\triangle ABC$, $DE \parallel BC$, $AD = 4$ cm, $AB = 9$ cm and $BC = 18$ cm. Find $DE$. **2**

**14.** The areas of two similar triangles are $20$ cm² and $245$ cm². If a side of the first is 2 cm, find the corresponding side of the second. **2**

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at $O$. If $AB = 5$ cm and $DC = 20$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. **3**

**16.** Find the altitude of an equilateral triangle of side 10 cm, in terms of $\sqrt{3}$. **3**

**17.** $\triangle ABC$ is right-angled at $B$ with $AB = 8$ cm and $BC = 15$ cm; $BD \perp AC$. Find $AD$, $DC$ and $BD$. **3**

**OR**

In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at $O$. If $AB = 6$ cm and $DC = 21$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. **3**

**18.** Find the altitude of an equilateral triangle of side 12 cm, in terms of $\sqrt{3}$. **3**

**OR**

$\triangle ABC$ is right-angled at $B$ with $AB = 7$ cm and $BC = 24$ cm; $BD \perp AC$. Find $AD$, $DC$ and $BD$. **3**

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** A triangular plot ABC has a straight fence DE with D on AB and E on AC such that $DE \parallel BC$. It is given that $AD = 6$ m, $DB = 4$ m and $AE = 9$ m. **5**

(i) State the theorem used to relate the segments.

(ii) Find EC.

(iii) Find the lengths AB and AC.

(iv) If $BC = 20$ m, find DE, and the ratio of the areas of $\triangle ADE$ and $\triangle ABC$.

**20.** Two ships leave a port at the same time; one sails 90 km due north and the other 120 km due west. Find the distance between them and the ratio of this distance to the one if both had sailed only half as far. **5**

**OR**

A girl 3 m tall walks away from a lamp post 9 m high at 2 m/s. Find the length of her shadow after 6 seconds. **5**

## END OF QUESTION PAPER — Total: 40 Marks